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School of Information Technology

Department of Computer Science

MIT 805: Big Data Semester Project (2026)

Part 2: MapReduce & Visualization

Makhosazane Mthethwa (u14241839) | Pumlisa Lusiba (u25162323)

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Prepared for: Dr. Olaperi Okuboyejo

GitHub Repository: [Pumlisa97/mit805-airline-delays-pyspark: MIT 805 Big Data project: U.S. airline on-time performance analysis using PySpark](#)

Arrival-delay risk by scheduled departure time, 2020-2024

1. Processing objective

Central Question: What is the relationship between the scheduled departure hour and the likelihood of a completed flight arriving at least 15 minutes late during the period from 2020 to 2024? Furthermore, how does the distribution of delay minutes attributed to a late-arriving aircraft vary by departure period? The aim of this analysis is to evaluate year-hour reliability, compare operational efficiencies during early morning, midday, and evening periods, and translate the observed differences into effective scheduling and disruption-recovery strategies. It should be noted that this analysis is of a descriptive character rather than a causal investigation.

2. MapReduce-style implementation

This analysis uses data from the U.S. Bureau of Transportation Statistics (BTS). It includes records of on-time performance for domestic non-stop flights. The full dataset contains 120 monthly CSV files from January 2015 to December 2024, totalling about 26.495 GiB of uncompressed data (approximately 28.45 GB). The downloadable ZIP collection is 2.971 GiB.

The source is the BTS TranStats Reporting Carrier On-Time Performance (1987–present) table, which provides a public download interface and definitions for the fields used here. The analysis uses these publicly accessible records for academic research, credits BTS as the data source, and directs readers to the original download page rather than redistributing the raw files in GitHub. U.S. federal government works are generally in the public domain under U.S. law, but we do not claim that BTS has assigned a specific Creative Commons license to this table.

For this project, we focused on a working dataset of 60 monthly CSV files from January 2020 to December 2024, which amounts to 14.130 GB (or 13.159 GiB). We checked and confirmed all 60 months as complete. This five-year subset meets processing requirements while being manageable in Google Colab. The working dataset served as a representative subset of the larger 2015-2024 dataset and was used for all PySpark-based processing. Data from 2020 reflects a disrupted baseline due to the impact of the COVID-19 pandemic on aviation operations.

The key fields used in this analysis are **Year**, **FlightDate**, **CRSDepTime**, **Cancelled**, **Diverted**, **ArrDelay**, and the five cause-of-delay columns. The main measure we used is completed arrivals, defined as flights that were not cancelled or diverted and that have a non-null ArrDelay. A late completed arrival has an ArrDelay of 15 minutes or more. After validation and Spark aggregation, we found 31,339,835 records with valid scheduled times, including 5,655,716 late completed arrivals. The total includes cancelled and diverted flights, so it should not be used as the measure for the primary calculation.

Question to measure mapping

Question component	Measure/analytical key
Hourly reliability	$\text{sum}(\text{LateCompletedArrival}) / \text{sum}(\text{CompletedArrival})$, keyed by Year and ScheduledDepHour
Operational Periods	05:00-09:59 early morning; 10:00-15:59 midday; 16:00-22:59 evening
Propagation Proxy	$\text{LateAircraftDelay minutes} / \text{sum of five attributed cause-minute categories among late completed arrivals}$

3. PySpark mapping and transformations

The notebook processes a list of 60 validated file paths using Spark to read CSV files. It uses options to include headers, not infer schema, and allow for some errors during reading. The notebook checks for necessary columns and changes the Year and scheduled time to integers. It also converts cancellation, diversion, and delay fields to doubles, and it parses FlightDate. It filters the data for the years 2020 to 2024 and ensures the dates and times are valid (with times in HHMM format from 0 to 2359, and minutes below 60). The main transformations used are select, filter, and withColumn. These transformations create new columns, such as ScheduledDepHour, which is calculated by taking the floor of CRSDepTime divided by 100, and also derive DeparturePeriod, CompletedArrival, and LateCompletedArrival.

```
flights_mapped = (raw_flights.select(...typed columns...)
    .filter(F.col("Year").between(2020, 2024))
    .filter(F.col("FlightDate").isNotNull())
    .filter(valid_scheduled_time)
    .withColumn("ScheduledDepHour",
        F.floor(F.col("CRSDepTime") / 100).cast("int"))
    .withColumn("CompletedArrival", F.when(
        (F.col("Cancelled") == 0) & (F.col("Diverted") == 0)
        & F.col("ArrDelay").isNotNull(), 1).otherwise(0))
    .withColumn("LateCompletedArrival", F.when(
        (F.col("CompletedArrival") == 1)
        & (F.col("ArrDelay") >= 15), 1).otherwise(0)))
```

The code checks if ArrDel15 exists in the source but defines late arrival as ArrDelay being 15 minutes or more. This makes the threshold clear. There are six excluded hours: 23:00 and 00:00–04:59. The displayed periods cover 05:00–22:59, or 18 of 24 hours. Therefore, the period rates should not cover all scheduled flights. Filtering for valid hours and dates also reduces the number of flights analyzed.

4. Shuffle/group/reduce operation

This analysis groups data by year and scheduled departure hour to calculate several metrics, including the number of scheduled flights, cancellations, diversions, arrivals that were completed, late arrivals, total minutes of delays for late arrivals, and totals for five cause minutes. Cause minutes are counted only when a flight is marked as late, treating any missing cause values as zero. Then, a smaller Spark aggregation groups the 120 hourly records into 15 periods based on year and

departure period. Only the reduced data was used for plotting; no detailed flight-level data was collected in Pandas.

```
hourly_by_year = (flights_mapped
    .groupBy("Year", "ScheduledDepHour")
    .agg(F.count("*").alias("scheduled_flights"),
        F.sum("CompletedArrival").alias("completed_arrivals"),
        F.sum("LateCompletedArrival").alias("late_completed_arrivals"),
        ... five conditional cause-minute sums ...)
    .withColumn("late_rate_completed", F.when(
        F.col("completed_arrivals") > 0,
        F.col("late_completed_arrivals") /
        F.col("completed_arrivals"))))
```

5. Distributed execution evidence

The transformations defined a lazy `DataFrame` plan; `hourly_by_year.collect()` triggered its execution and returned 120 aggregated rows. We kept a detailed plan of how this worked, showing the steps: Scan CSV, Filter, Project, partial HashAggregate, hash-partitioning Exchange, final HashAggregate, then a range-partitioning Exchange and Sort.

The first exchange redistributes partial results based on the grouping keys, so matching keys go to the same final aggregator. The second exchange is set up to manage the sorting order, rather than the analytical grouping.

```
CSV scan      -> filter/project -> partial HashAggregate
              -> Exchange hashpartitioning(Year, ScheduledDepHour, 200)
              -> final HashAggregate -> rate calculation
              -> Exchange rangepartitioning(Year, ScheduledDepHour, 200)
              -> Sort -> collect(120 rows)
```

The excerpt below discusses a saved operator from the formatted plan. The number 200 refers to the planned shuffle partition count, not the actual number of completed tasks. The plan is labeled as `AdaptiveSparkPlan` and is marked as not being the final version. This plan allows for distributed operations, but without access to the Spark UI, we cannot see the stage timings, completed task counts, or how well the runtime is parallelized. The application ID started with "local-", indicating that this run was executed in Spark's local mode instead of using a multi-node cluster.

(6) HashAggregate Keys: [Year, ScheduledDepHour]

Functions: partial_count, partial_sum(...)

(7) Exchange hashpartitioning(Year, ScheduledDepHour, 200)

(8) HashAggregate Keys: [Year, ScheduledDepHour]

Functions: count, sum(...)

(10) Exchange rangepartitioning(Year, ScheduledDepHour, 200)

(11) Sort Year ASC, ScheduledDepHour ASC

(12) AdaptiveSparkPlan isFinalPlan=false

In classic Hadoop MapReduce, a mapper produces key-value pairs. Then, a shuffle organizes these keys across different workers, and a reducer writes the grouped output, usually using disk storage. In contrast, Spark creates and improves a Directed Acyclic Graph (DAG) of transformations on DataFrames. It can streamline narrow operations, while exchanges define shuffle boundaries. Here, the HashAggregate step before the first exchange combines records locally, reducing the amount of data that needs to be shuffled. The saved plan serves as evidence of execution but does not prove that it's happening in a distributed cluster.

6. Results: hourly and annual reliability

Annual rates are calculated using completed arrivals as the denominator. These rates increased from 9.82% in 2020 to 21.08% in 2022 and stabilized between 20% and 21% in 2023 and 2024. However, because flight volumes varied and the pandemic created exceptional circumstances, a direct comparison of data from 2020 to 2024 is not suitable for establishing causal relationships.

Year	Scheduled	Completed	Late completed	Late rate
2020	4,688,354	4,399,575	431,921	9.82%
2021	5,995,397	5,878,219	1,010,332	17.19%
2022	6,729,125	6,532,012	1,376,798	21.08%
2023	6,847,899	6,743,403	1,386,699	20.56%
2024	7,079,060	6,965,246	1,449,966	20.82%

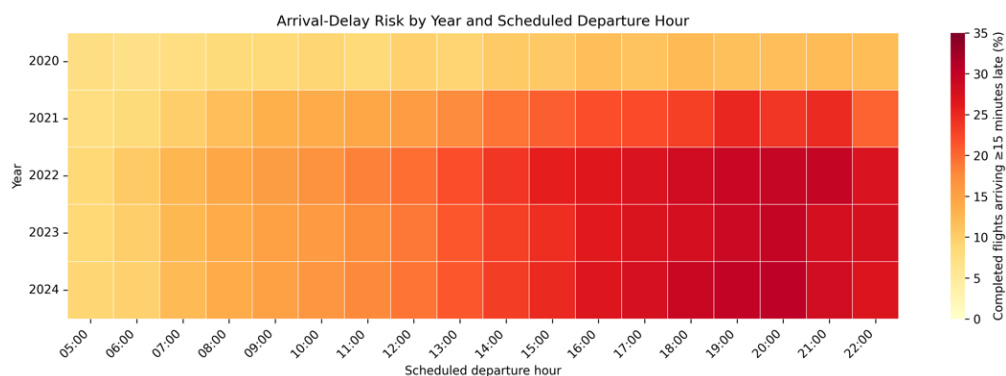


Figure 1: Completed-flight late-arrival rate by scheduled departure hour and year; darker cells indicate greater risk. The heatmap displays 05:00-22:00, whereas the 120-row aggregation covers all 24 hours across five years; the figure displays 18 hours per year (05:00–22:00). The six excluded hours account for the difference.

At 05:00 in 2024, 9.06% of completed arrivals were at least 15 minutes late (17,856 of 197,159). At 19:00, the corresponding rate was 29.98% (113,930 of 380,007), a difference of 20.92 percentage

points. Comparing rates at different times is more useful for scheduling than a single overall rate. However, differences in routes, airlines, airports, and flight volumes can complicate this.

7. Period results and attributed causes

The shorter period calculates rates using total counts instead of averaging hourly percentages. It sums five reported delay-cause categories for late completed arrivals, then calculates late-aircraft minutes as a percentage of their combined attributed minutes. These methods help ensure the correct weighting and totals.

2024 period	Scheduled	Late completed	Late rate	Late aircraft share
Early morning	2,067,230	250,135	12.27%	21.44%
Midday	2,538,571	510,461	20.40%	41.58%
Evening	2,394,989	673,211	28.66%	46.89%

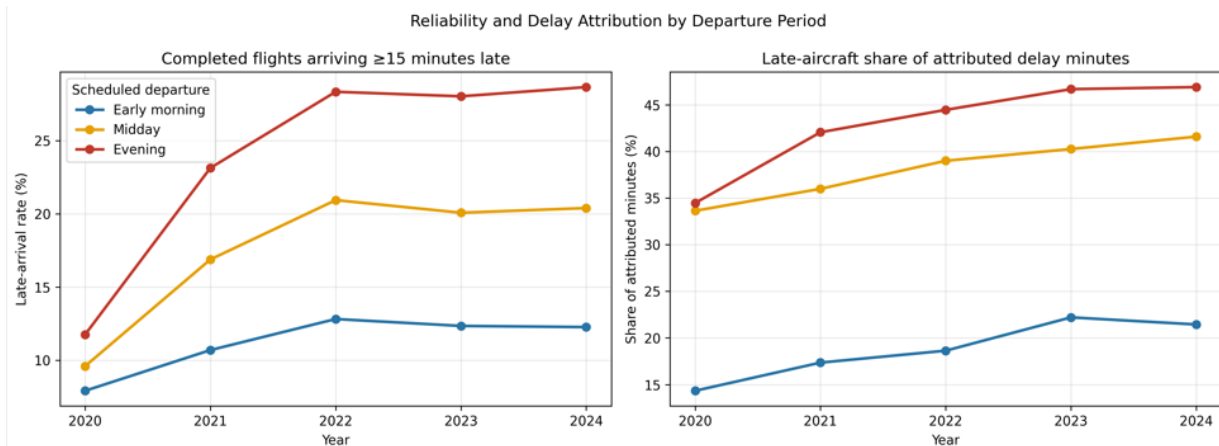


Figure 2: Late-arrival rates (left) and late-aircraft share of five attributed delay-minute categories(right), for three scheduled departure periods in 2020-2024. The share is of attributed minutes, not the share of flights caused by late aircraft.

In 2024, the rate of completed flights during the evening surpassed that of the early morning by 16.39 percentage points. Specifically, the share of late aircraft in the evening was 46.89%, compared to only 21.44% during the early morning hours. Analyzing the overall data, late aircraft and carrier-attributed minutes accounted for 146,499,648 (37.99%) and 145,996,458 (37.86%) of the total 385,650,737 attributed minutes. Notably, the total minutes attributed to causes differed from the total minutes associated with late arrivals by 1,120 minutes. These figures represent recorded attribution totals and do not serve as independent evidence of a delay propagation mechanism.

8. Business/societal interpretation

For airline operations teams, the identification of time windows between evening and morning flights in 2024 presents an opportunity to review various operational aspects, including turnaround buffers, the availability of spare aircraft, and recovery plans. This hour-by-hour reliability pattern can be beneficial for airports and passengers when planning connection windows or assessing the risk of disruptions.

A logical next step would be to analyze data stratified by airport, route, and carrier before making any changes to flight schedules. It's important to note that the current results highlight associations rather than the direct effects of rescheduling flights.

Dataset and analytical limitations.

- First, this data only covers direct domestic flights from certain airlines; it does not include all airlines, international trips, or complete connecting flights.
- Second, the completed-arrival rate only counts flights that landed as planned, excluding canceled or diverted flights and those without a recorded delay. So, it does not show the full likelihood that a passenger will reach their destination as intended.
- Third, the cause-minute fields list categories for delayed flights. The late-aircraft share shows how many minutes are attributed to this issue, not how many flights were actually delayed by a late aircraft. Missing cause-minute entries are counted as zero in this calculation.
- Fourth, the disruptions in 2020 and changes in the mix of routes, airports, and airlines may affect the differences we see. Thus, the 2024 morning-to-evening gap should help us find areas to investigate, rather than being seen as an effect of changing departure times.

Reference

1. Bureau of Transportation Statistics (BTS) (2021) Full year 2020 and December 2020 U.S. airline traffic data. U.S. Department of Transportation. Available at: <https://www.bts.gov/newsroom/full-year-2020-and-december-2020-us-airline-traffic-data> (Accessed: 27 August 2026).
2. Bureau of Transportation Statistics (BTS) (2026a) Reporting Carrier On-Time Performance (1987–Present): Download data. TranStats, U.S. Department of Transportation. Available at: https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ (Accessed: 27 August 2026).
3. Bureau of Transportation Statistics (BTS) (2026b) Reporting Carrier On-Time Performance (1987–Present): Database information. TranStats, U.S. Department of Transportation. Available at: https://www.transtats.bts.gov/DatabaseInfo.asp?QO_VQ=EFD (Accessed: 27 August 2026).
4. Bureau of Transportation Statistics (BTS) (2026c) Technical Directive No. 40: Reporting on-time performance. U.S. Department of Transportation. Available at: <https://www.bts.gov/explore-topics-and-geography/modes/aviation/number-40-technical-directive-reporting-time> (Accessed: 27 August 2026).

Appendix A: Supplementary Airline-Level Analysis (2024)

The primary analysis in this study investigates how scheduled departure time influences the likelihood of a completed flight arriving at least 15 minutes late between 2020 and 2024. The supplementary analyses presented in this appendix provide additional operational context by examining airline-level performance, seasonal delay patterns, and delay-cause distributions during 2024. These findings complement the main analysis but do not form the primary analytical focus of the study.

Figure A1: Airline Performance and Monthly Delay Trends

